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Lighting fixture

Abstract

The present disclosure provides a lighting fixture. The lighting fixture includes: a light emitting plate, a rotating shaft and a light transmitting plate. The rotating shaft is mounted on a light exiting side of the light emitting plate. The light transmitting plate is located on the light exiting side of the light emitting plate and is rotatably connected to the rotating shaft. Light emitted from the light emitting plate comes out after passing through the light transmitting plate. When the light transmitting plate is driven to make a rotating movement around a center axis of the rotating shaft, a distance between the light transmitting plate and the light emitting plate changes, and light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out changes accordingly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority under 35 U.S.C. § 119 (a) to Chinese Patent Application No. 202323532086.6, filed Dec. 22, 2022. The entire disclosure of Chinese Patent Application No. 202323532086.6 is hereby incorporated by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a field of lighting, and in particular to a lighting fixture.

BACKGROUND

(3) In the prior art, a light exiting angle of a lamp is generally fixed. Of course, there are some lamps that can adjust their light exiting angles, but an adjustment structure of the light exiting angle of these lamps is complicated, has many parts, is costly, is not easy to operate, and causes inconvenience to the user.

SUMMARY

(4) In view of this, the present disclosure provides a lighting fixture, which can solve the above technical problems.

(5) The present disclosure provides a lighting fixture. The lighting fixture includes: a lighting emitting plate, a rotating shaft, and a light transmitting plate. The rotating shaft is mounted on a light exiting side of the lighting emitting plate. The light transmitting plate is located on the light exiting side of the lighting emitting plate, and is rotatably connected to the rotating shaft. Light emitted from the light emitting plate comes out after passing through the light transmitting plate. When the light transmitting plate is driven to make a rotating movement around a center axis of the rotating shaft, a distance between the light transmitting plate and the light emitting plate changes,

and a light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out changes accordingly.

(6) In some possible embodiments of the present disclosure, a circumferential wall of the rotating shaft defines a plurality of spiral grooves; the lighting fixture includes a plurality of plug elements; the plurality of plug elements are in one-to-one correspondence with the plurality of spiral grooves; each of the plurality of plug elements includes a first end and a second end set opposite to the first end, the first end of each plug element is connected to the light transmitting plate, the second end of each plug element is inserted into the corresponding spiral groove and is capable of sliding within the spiral groove following the rotating movement of the light transmitting plate.

(7) In some possible embodiments of the present disclosure, the spiral groove has at least two limiting points within the spiral groove; the at least two limiting points are spaced apart within the spiral groove along a spiral direction of the spiral groove; when the plug element is clamped to different limiting points within the spiral groove, the distance between the light transmitting plate and the light emitting plate is different, and the light which is emitted from the light emitting plate and then passes through the light transmitting plate comes out at different light exiting angles.

(8) In some possible embodiments of the present disclosure, the spiral groove includes a spiral portion, the at least two limiting points are defined within the spiral portion.

(9) In some possible embodiments of the present disclosure, the spiral groove further includes a transition portion and an insertion portion; the transition portion interconnects the spiral portion with the insertion portion; the rotating shaft further includes an end face at a side away from the light emitting plate; and the insertion portion extends from the circumferential wall of the rotating shaft and penetrates through the end face of the rotating shaft.

(10) In some possible embodiments of the present disclosure, an opening of the insertion portion in the end face of the rotating shaft is larger than an outer diameter of the second end of the plug element.

(11) In some possible embodiments of the present disclosure, the spiral portion includes a first sidewall surface close to the end face; the spiral portion includes a stop portion formed on an end of the spiral portion close to the transition portion; the stop portion closes to the opening of the insertion portion and protrudes from the first sidewall surface toward a side away from the end face.

(12) In some possible embodiments of the present disclosure, the spiral portion includes a first sidewall surface close to the end face, the spiral portion further includes a second sidewall surface at a side away from the end face; the second end of the plug element includes a convex ring; and the convex ring is mated with the first sidewall surface and the second sidewall surface of the spiral portion, respectively.

(13) In some possible embodiments of the present disclosure, the light transmitting plate further includes a light transmitting portion and a mounting portion; the light transmitting portion surrounds the mounting portion; the mounting portion is recessed relative to the light transmitting portion to form a recess on a side away from the light emitting plate; an inner side wall of the recess defines a plurality of mounting holes; the first ends of the plurality of plug elements are connected within the plurality of mounting holes, respectively.

(14) In some possible embodiments of the present disclosure, the lighting fixture further includes a limiting member and a heat sink, the heat sink includes a circumferential sidewall and a mounting plate; the circumferential sidewall surrounds the mounting plate to form a mounting groove; the light emitting plate and the light transmitting plate are mounted within the mounting groove; the light transmitting plate is located on a side of the light emitting plate away from the heat sink; the limiting member is located on a side of the light transmitting plate away from the light emitting plate; one end of the limiting member is connected to the circumferential sidewall, and the other end of the limiting member is resting against a surface of the light transmitting plate away from the light emitting plate.

(15) In some possible embodiments of the present disclosure, the light transmitting plate includes a limiting portion at an edge of a side of the light transmitting plate away from the light emitting plate; the limiting portion cooperates with the limiting member to limit the rotating movement of the light transmitting plate relative to the light emitting plate.
(16) In some possible embodiments of the present disclosure, the mounting portion includes a connection port; the light fixture further includes a communication module; the communication module is mounted on the connection port.
(17) Therefore, in the present disclosure, the light which is emitted from the light emitting plate and then passes through the light transmitting plate comes out. A magnitude of a distance between the light transmitting plate and the light emitting plate is related to a light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out. When the distance between the light transmitting plate and the light emitting plate changes, the light exiting angle of the light which is emitted by the light emitting plate and passes through the light transmitting plate coming out is changed accordingly. Therefore, by rotating the light transmitting plate to adjust the distance between the light transmitting plate and the light emitting plate, the light exiting angle of the light which is emitted from the light emitting plate coming out and then passes through the light transmitting plate can be adjusted, which can simplify the structure, reduce the parts, reduce the costs, and facilitate the operations of the user.

Description

BRIEF DESCRIPTION OF THE ACCOMPANYING DRAWINGS
(1) To describe the technology solutions in the embodiments of the present disclosure more clearly, the following briefly introduces the accompanying drawings required for describing the embodiments. Obviously, the accompanying drawings in the following description show merely at least one embodiment of the present disclosure, those of ordinary skilled in the art may also obtain other drawings based on these drawings without any creative efforts.
(2) FIG. **1** illustrates a schematic view of a three-dimensional structure of a lighting fixture in accordance with one embodiment of the present disclosure.
(3) FIG. **2** illustrates an exploded view of FIG. **1**.
(4) FIG. **3** illustrates a cross-sectional view of FIG. **2** taken along a line III-III.
(5) FIG. **4** illustrates a partial enlarged view of FIG. **3** at IV.
(6) FIG. **5** illustrates a schematic view of an assembly of a rotating shaft and a plug element in accordance with one embodiment of the present disclosure.
(7) FIG. **6** illustrates an exploded view of FIG. **5**.
(8) FIG. **7** illustrates a schematic view of a three-dimensional structure of the lighting fixture of FIG. **1** from another view.
(9) FIG. **8** illustrates an exploded view of FIG. **7**.
(10) FIG. **9** illustrates an enlarged view of a light transmitting member of FIG. **2**.
(11) FIG. **10** illustrates a partially enlarged view of FIG. **9** at IX.
(12) FIG. **11** illustrates a top view of FIG. **7**.
(13) FIG. **12** illustrates a partially enlarged view of FIG. **11** at XI.
DETAILED DESCRIPTION OF ILLUSTRATED EMBODIMENTS
(14) Embodiments of the present disclosure are described in detail below, and examples of the embodiments are shown in the accompanying drawings, wherein the same or similar labeling throughout denotes the same or similar elements or elements having the same or similar functions.
(15) The embodiments described below with reference to the accompanying drawings are exemplary and are intended to be used to explain the present disclosure and are not to be construed as a limitation of the present disclosure.

(16) It should be noted that all directional indications (e.g., up, down, left, right, front, back . . .) in the embodiments of the present disclosure are only used to explain the relative positional relationship, movement and the like among the components in a particular state (as shown in the accompanying drawings), and that the directional indications are correspondingly changed if the particular state is changed.
(17) Furthermore, the terms “first” and “second” are used only for descriptive purposes, and are not to be construed as indicating or implying relative importance or implicitly specifying the number of technical features indicated. Thus, a feature defined with “first” or “second” may include one or more such features, either explicitly or implicitly. In the description of the present disclosure, “plurality of” means two or more, unless otherwise expressly and specifically limited.
(18) In this application, unless otherwise expressly specified and limited, the terms “connected”, “fixed”, etc. are to be understood broadly, e.g., they may be fixedly connected, removably connected, or integrally connected; they may be mechanically connected or electrically connected; they may be directly connected or indirectly connected through an intermediate medium, a connection within two elements or an interactive relationship between two elements. For those of ordinary skill in the art, the specific meanings of the above terms in this application may be understood on a case-by-case basis.
(19) Referring to FIGS. **1** to **4**, FIG. **1** illustrates a schematic view of a three-dimensional structure of a lighting fixture **1** in accordance with one embodiment of the present disclosure; FIG. **2** illustrates an exploded view of FIG. **1**; FIG. **3** illustrates a cross-sectional view of FIG. **2** taken along a line III-III; FIG. **4** illustrates a partial enlarged view of FIG. **3** at IV.
(20) As shown in FIG. **2**, the lighting fixture **1** includes a lighting emitting plate **11**, a rotating shaft **12**, and a light transmitting plate **13**. The light emitting plate **11** includes a light exiting side **111**. The light exiting side **111** is a side of the light emitting plate **11** close to the light transmitting plate **13**. As shown in FIGS. **3** and **4**, the rotating shaft **12** is mounted on the light exiting side **111** of the lighting emitting plate **11**. The light transmitting plate **13** is located on the light exiting side **111** of the lighting emitting plate **111**, and is rotatably connected to the rotating shaft **12**. Light emitted from the light emitting plate **11** passes through the light transmitting plate **13** and comes out. When the light transmitting plate **13** is driven to make a rotating movement around a center axis of the rotating shaft **12**, a distance between the light transmitting plate **13** and the light emitting plate **11** is changed, and a light exiting angle of the light which is emitted from the light emitting plate **11** and passes through the light transmitting plate **13** coming out changes accordingly. The light exiting angle is an angle A at which all light emitted from the light emitting plate **11** is distributed after coming out the light transmitting plate **13**.
(21) Therefore, in this application, the light which is emitted from the light emitting plate **11** and passes through the light transmitting plate **13** comes out. A magnitude of the distance between the light transmitting plate **13** and the light emitting plate **11** is related to a light exiting angle of the light which is emitted from the light emitting plate **11** and then passes through the light transmitting plate **13** coming out. When the distance between the light transmitting plate **13** and the light emitting plate **11** changes, the light exiting angle of the light which is emitted by the light emitting plate **11** and then passes through the light transmitting plate **13** coming out is changed accordingly. Therefore, by rotating the light transmitting plate **13** to adjust the distance between the light transmitting plate **13** and the light emitting plate **11**, the light exiting angle of the light which is emitted from the light emitting plate **11** and passes through the light transmitting plate **13** coming out can be adjusted, which can simplify the structure, reduce the parts, reduce the costs, and facilitate the operations of the user.
(22) In some embodiments, the light emitting plate **11** may emit light. For example, the light emitting plate **11** is mounted with at least one light strip on one side of the light emitting plate **11** towards the light transmitting plate **13**; the light strip may be, but is not limited to, an LED light strip; the light strip is a plurality of light strip circles spreading from a center of the light

transmitting plate **11** towards an edge of the light transmitting plate **11**. In another embodiment, the light strip may be arranged in a zigzag pattern, etc. The light strip may emit light. In other embodiments, the light emitting plate **11** may include a plurality of light beads on a side of the light emitting plate **11** towards the light transmitting plate **13**. The light beads may emit light and are not limited herein.

(23) Therefore, the light emitting plate **11** is capable of emitting light and the emitted light comes out after passing through the light transmitting plate **13**.

(24) In some embodiments, referring again to FIGS. **2** and **3**, the light transmitting plate **13** is of semi-transparent plastic or a semi-transparent glass structure. The light transmitting plate **13** protrudes outwardly towards one side away from the light emitting plate **11**. The light transmitting plate **13** spreads outwardly from a center of the light transmitting plate **13** in a uniform arrangement to form a plurality of tubular light transmission circles **131**. A depth of depression of each light transmission circle **131** close to one side of the light emitting plate **11** is 3˜5 mm. A diameter of the pipeline of each light transmission circle **131**, i.e., a radial width, is 6.4-10 mm. A distance between the centers of two adjacent light transmission circles **131** is 10-15 mm. It is understood that in other embodiments, the depth of depression of the light transmission circle **131** and the diameter of the light transmission circle **131** can be adjusted according to the actual needs.

(25) Therefore, the lighting fixture **1** of the present disclosure, as a result of using the light transmitting plate **13** with a plurality of light transmission circles **131** having the above-described parameters, such that its UGR (Unified Glare Rating) values are all less than 28, glare may be reduced.

(26) For some embodiments, please referring to FIGS. **5** and **6** together, FIG. **5** illustrates a schematic view of an assembly of a rotating shaft **12** and a plug element **14** in accordance with one embodiment of the present disclosure; FIG. **6** illustrates an exploded view of FIG. **5**. The rotating shaft **12** is cylindrical in shape. The rotating shaft **12** has a circumferential wall **121**. The circumferential wall **121** defines a plurality of spiral grooves **122**. The plurality of spiral grooves **122** are spiral grooves having a center axis of the rotating shaft **12** as a rotating axis. The lighting fixture **1** includes a plurality of plug elements **14**. The plurality of plug elements **14** are in one-to-one correspondence with the plurality of spiral grooves **122**. Each plug element **14** includes a first end **141** and a second end **142** opposite to the first end **141**. The first end **141** of each plug element **14** is connected to the light transmitting plate **13**, and the second end **142** of each plug element **14** is inserted into the corresponding spiral groove **122**. The plug element **14** is capable of sliding within the spiral groove **122** following a rotating movement of the light transmitting plate **13**.

(27) Therefore, through the cooperation of the plug element **14** with the spiral groove **122** on the rotating shaft **12**, the rotating movement of the light transmitting plate **13** relative to the rotating shaft **12** can be more stable and reliable, with better stroke stability.

(28) In some embodiments, referring again to FIGS. **5** and **6**, the plurality of spiral grooves **122** are spaced apart in a circumferential direction of the circumferential wall **121** of the rotating shaft **12**. In this embodiment, the plurality of spiral grooves **122** includes three spiral grooves **122**, and the three spiral grooves **122** are spaced apart in the circumferential direction of the circumferential wall **121** of the rotating shaft **12**. It can be appreciated that the three spiral grooves **122** have the same structural shape and are simply distributed at different locations in the circumferential direction of the circumferential wall **121** of the rotating shaft **12**. In other embodiments, the number of the spiral grooves **122** is not limited to three, but may be one, two, four or more than four, without limitation herein.

(29) In some embodiments, referring again to FIGS. **5** and **6**, the spiral groove **122** includes at least two limiting points **1221**. The at least two limiting points **1221** are spaced apart within the spiral groove **122** along a spiral direction of the spiral groove **122**. When the plug element **14** engages with different limiting points **1221** within the spiral groove **122**, the light transmitting plate **13** and the light emitting plate **11** are at different distances, and the light which is emitted by the light

emitting plate **11** and then passes through the light transmitting plate **13** comes out at different light exiting angles. In this embodiment, the at least two limiting points **1221** include three limiting points **1221**, and the three limiting points **1221** are spaced apart within the spiral groove **122** along the spiral direction of the spiral groove **122**. Thus, the light which is emitted by the light emitting plate **11** and then passes through the light transmitting plate **13** comes out at three light exiting angles, e.g., 60°, 85° and 110° respectively. In other embodiments, the at least two limiting points **1221** include, but are not limited to, three limiting points **1221**, and the light which is emitted by the light emitting plate **11** and then passes through the light transmitting plate **13** comes out at a light exiting angle including, but is not limited to, 60°, 85° and 110°, without limitation herein.
(30) Therefore, in sliding in the spiral groove **122** following the rotating movement of the light transmitting plate **13**, the plug element **14** may be engaged with different limiting points **1221** within the spiral groove **122** to define the light emitting plate **11** relative to the light transmitting plate **13** at different light exiting angles.
(31) In some embodiments, the limiting point **1221** is a recess, and the plug element **14** is an elastic pin.
(32) Therefore, when the plug element **14** slides within the spiral groove **122** but does not slide to a position at the current limiting point **1221**, the plug element **14** is in an elastic compression state. When the plug element **14** slides within the spiral groove **122** to a position at the current limiting point **1221**, the plug element **14** extends under its elastic restoring force into the current limiting point **1221** and engages with the current limiting point **1221**. When it is necessary to adjust the light exiting angle of the light emitting plate **11** relative to the light transmitting plate **13**, the light transmitting plate **13** is rotated, the light transmitting plate **13** drives the plug element **14** to move, causing the plug element **14** to be compressed and retracted to overcome the resistance generated by the current limiting point **1221**, and forcing the plug element **14** to detach from the current limiting point **1221** and to continue moving to a next limiting point **1221** and engage with the next limiting point **1221**.
(33) In some embodiments, an inner wall surface of the recess of the limiting point **1221** is a wedge-shaped surface. Therefore, an end of the second end **142** of the plug element **14** is also wedge-shaped.
(34) Therefore, since the inner wall surface of the recess of the limiting point **1221** is a wedge-shaped surface and the end of the second end **142** of the plug element **14** is also wedge-shaped, the plug element **14** is able to be inserted into the limiting point **1221** or out of the limiting point **1221** much more easily by the fit of the wedge-shaped surface and the wedge-shaped end.
(35) In some embodiments, referring again to FIGS. **5** and **6**, the spiral groove **122** includes a spiral portion **1222**, a transition portion **1223** and an insertion portion **1224**. The at least two limiting points **1221** are defined within the spiral portion **1222**. The transition portion **1223** interconnects the spiral portion **1222** with the insertion portion **1224**. The rotating shaft **12** further includes an end face **1220** at a side away from the light emitting plate **11**. The insertion portion **1224** extends from a circumferential wall **121** of the rotating shaft **12** and penetrates through the end face **1220** of the rotating shaft **12**.
(36) Therefore, when the light transmitting plate **13** with the plug elements **14** mounted thereon is sleeved with the rotating shaft **12**, the second end **142** of the plug element **14** can slide through the insertion portion **1224** into the spiral portion **1222**; after the second end **142** of the plug element **14** has been inserted into the insertion portion **1224**, the light transmitting plate **13** is driven to rotate, the light transmitting plate **13** drives the plug element **14** to run through the transition portion **1224**, and slides into the spiral portion **1222**, simplifying an operation of sleeve the light transmitting plate **13** with the plug element **14** to the rotating shaft **12**.
(37) In some embodiments, the insertion portion **1224** has an opening dimension on the end face **1220** of the rotating shaft **12** that is larger than an outer diameter of the second end **142** of the plug element **14**.

(38) Therefore, since the opening dimension of the insertion portion **1224** on the end face **1220** of the rotating shaft **12** is larger than the outer diameter of the second end **142** of the plug element **14**, the second end **142** of the plug element **14** can be inserted into the insertion portion **1224** more easily, further simplifying the process of insertion operation of inserting the light transmitting plate **13** with the plug elements **14** mounted on the rotating shaft **12**.
(39) In some embodiments, referring again to FIGS. **5** and **6**, the spiral portion **1222** includes a first sidewall surface **1227** close to the end face **1220**. The spiral portion **1222** has a stop portion **1226** formed on an end close to the transition portion **1223**. The stop portion **1226** is adjacent to an opening of the insertion portion **1224** and protruded from the first sidewall surface **1227** towards a direction away from the end face **1220**.
(40) Therefore, when the plug element **14** slides within the spiral portion **1222**, the stop portion **1226** provides a limit to the sliding of the plug element **14** within the spiral portion **1222**, avoiding that during normal adjustment of the light exiting angle of the light transmitting plate **13** relative to the light emitting plate **11**, the plug element **14** slips out of the spiral portion **1222**, which causes the light transmitting plate **13** to separate from the light emitting plate **11**, thereby causing operational inconvenience to the user.
(41) In some embodiments, the spiral portion **1222** includes a second sidewall surface **1228** on a side away from the end face **1220**, and in some embodiments, the distance between the first sidewall surface **1227** and the second sidewall surface **1228** of the spiral portion **1222** is approximately equal to a maximum outer diameter of the plug element **14**.
(42) Therefore, the spiral portion **1222** may provide a stable and reliable guiding action for the plug element **14**.
(43) In some embodiments, the second end **142** of the plug element **14** includes a convex ring **143**. The convex ring **143** is configured to cooperate with the first sidewall surface **1227** and the second sidewall surface **1228** of the spiral portion **1222**, respectively. In this way, the plug element **14** other than the convex ring **143** can have a smaller outer diameter size relative to the convex ring **143**, which can save the cost of material used for the plug element **14** and reduce the overall weight of the lighting fixture **1**.
(44) In some embodiments, the insertion section **1224** includes a third sidewall surface **1229** at a side away from the end face **1220**. The second sidewall surface **1228** and the third sidewall surface **1229** are substantially flush. A fourth sidewall surface **1230** of the transition section **1223** on the side away from the end face **1220** is recessed relative to the second sidewall surface **1228** and the third sidewall surface **1229** toward the side away from the end face **1220**. As a result, the transition portion **1223** is substantially in a shape of a curved moon. The fourth sidewall surface **1230** is located face to face with the stop portion **1226**.
(45) Therefore, the transition portion **1223** itself has an effect of limiting the plug element **14**.
(46) In some embodiments, the rotating shaft **12** includes a plurality of first mounting portions **124** located on a side of the light emitting plate **11** close to the light emitting plate **11**. The rotating shaft **12** is mounted to the surface of the light exiting side **111** of the light emitting plate **11** through the plurality of first mounting portions **124**. In some specific embodiments, the plurality of the first mounting portions **124** are a plurality of lugs spaced apart and projecting outwardly from a side of a circumferential wall **121** of the rotating shaft **12** proximate the light emitting plate **11**. In other embodiments, the first mounting portions **124** may be connected to form a mounting ring. In other embodiments, the first mounting portions **124** may also be mounting holes defined on an end face of the rotating shaft **12** close to the light emitting plate **11**. No limitation is made herein.
(47) It can be appreciated that in some other embodiments, the transition portion **1223** and the insertion portion **1224** may be omitted.
(48) In some embodiments, referring to FIGS. **7** and **8** together, FIG. **7** illustrates a schematic view of a three-dimensional structure of the lighting fixture **1** of FIG. **1** from another view; FIG. **8** illustrates an exploded view of FIG. **7**. The light transmitting plate **13** includes a light transmitting

portion **1310** and a second mounting portion **132**. The light transmitting portion **1310** surrounds the second mounting portion **132**. The plurality of the light transmission circles **131** are located on the light transmission portion **1310**. Referring to FIGS. **9** and **10** together, the second mounting portion **132** is recessed towards a side away from the light emitting plate **11** relative to the light transmitting portion **1310** to form a mounting groove. The second mounting portion **132** is sleeved on the rotating shaft **12**. The second mounting portion **132** defines a plurality of mounting holes **133** formed by spaced recesses in an inner side wall of the second mounting portion **132**. An extension of a center axis of each the mounting hole **133** intersects with a center axis of the second mounting portion **132**. The first ends **14** of the plurality of plug elements **14** are mounted within the plurality of mounting holes **133** respectively.

(49) Therefore, the first ends **141** of the plurality of plug elements **14** can be reliably mounted within the plurality of mounting holes **133**, they are not easily dropped, simplifying mount and facilitating operation of the light transmitting plate **13**.

(50) In some embodiments, the mounting hole **133** includes a hole portion **1331** and an opening portion. The hole portion **1331** is cylindrical in shape. An extension of a center axis of the hole portion **1331** intersects or is perpendicular to the center axis of the second mounting portion **132**. The opening portion **1332** is disposed on a circumferential sidewall of the hole portion **1331** and is interconnected with the hole portion **1331**. The hole portion **1331** has a diameter dimension equal to a diameter dimension of the first end **141** of the plug element **14** so that the first end **141** of the plug element **14** can be mounted within the hole portion **1331**. The opening portion **1332** is located on the circumferential sidewall of the hole portion **1331** to facilitate molding as well as demolding of the mounting hole **133**.

(51) In some embodiments, referring again to FIG. **8**, on a side of the light transmitting plate **13** away from the light emitting plate **11**, the second mounting portion **132** is protruded relative to the light transmitting portion **1310** toward the side away from the light emitting plate **11**, and correspondingly, a position on the second mounting portion **132** corresponding to the mounting hole **133** is protruded in a radial direction away from the second mounting portion **132**, thereby forming a cylindrical and convex ribbed gripping portion **135**.

(52) When mounting or rotating the light transmitting plate **13**, the light transmitting plate **13** can be operated by holding the gripping portion **135**. Compared to operation of the light transmitting plate **13** by holding an edge of the light transmitting plate **13**, operation of the light transmitting plate **13** can be simplified by holding the gripping portion **135**.

(53) In some embodiments, referring again to FIG. **8**, the lighting fixture **1** further includes a limiting member **16** and a heat sink **18**. The heat sink **18** includes a circumferential sidewall **181** and a mounting plate **182**. The circumferential sidewall **181** is connected to a circumferential edge of the mounting plate **182** to form a mounting groove **183**. The light emitting plate **11** and the light transmitting plate **13** are mounted within the mounting groove **183**. The light transmitting plate **13** is located on a side of the light emitting plate **11** away from the heat sink **18**. Referring to FIGS. **11** and **12**, FIG. **11** illustrates a top view of FIG. **7**, FIG. **12** illustrates a partially enlarged view of FIG. **11** at XI. The limiting member **16** is located on a side of the light transmitting plate **13** away from the light emitting plate **11**. One end of the limiting member **16** is connected to the circumferential sidewall **181** and the other end of the limiting member **16** rests against a surface of a side of the light transmitting plate **13** away from the light transmitting plate **13**, to limit the light transmitting plate **13** from disengaging from the rotating shaft **12**.

(54) In some embodiments, the light transmitting plate **13** includes a limiting portion **136** at an edge of a side of the light transmitting plate **13** away from the light emitting plate **11**. In this embodiment, the limiting portion **136** is an indicator arrow for indicating a direction of rotation of the light transmitting plate **13**. The limiting portion **136** cooperates with the limiting member **16**, e.g., the limiting member **16** can be snapped in a middle position of the limiting portion **136**, and can limit the rotating movement of the light transmitting plate **13** relative to the light emitting plate

11.
(55) In some embodiments, referring to FIG. **11**, the light transmitting plate **13** includes a first marking portion **137** on a side of the light transmitting plate **13** away from the light emitting plate **11**. The circumferential sidewall **181** includes a second marking portion **185**. When the first marking portion **137** and the second marking portion **185** are aligned, the light transmitting plate **13** can be sleeved on the rotating shaft **12**.
(56) Therefore, with the first marking portion **137** and the second marking portion **185**, it becomes easier for the light transmitting plate **13** to be able to be sleeved on the rotating shaft **12**.
(57) In some embodiments, referring again to FIG. **8**, the heat sink **18** further includes an outer shell **186** and a plurality of heat dissipating fins **188**. The outer shell **186** is in a form of a column. The circumferential sidewall **181** is located on an inside of the outer shell **186**. The circumferential sidewall **181** is spaced apart from the outer shell **186**. The plurality of heat dissipating fins **188** are spaced apart from and connected to the outer shell **186** and the circumferential sidewall **181**. The plurality of heat dissipating fins **188** also extend on a side of the mounting plate **182** away from the light emitting plate **11**.
(58) Therefore, the outer shell **186** is a hollow annular ring, the mounting plate **182** has an outer diameter smaller than the inner diameter of the outer shell **186**, the circumferential sidewall **181** is located within the annular ring of the outer shell **186**, the circumferential sidewall **181** is spaced apart from the outer shell **186** to form a circular channel **189**, and each of the heat dissipating fins **188** is connected between the inner wall of the outer shell **186** and the circumferential sidewall **181** and extends to the mounting plate **11** on a side away from the light emitting plate **11**. The plurality of heat dissipating fins **188** are spaced apart in a circumferential direction of the mounting plate **182** with a center of the heat sink **18** being a center.
(59) Therefore, the heat generated by the light emitting plate **11** is conducted by the heat dissipating fins **188** to a portion of the heat dissipating fins **188** located within the circular channel **189**, which has a vertically sloping upward space equivalent to a chimney, thus enabling the use of the chimney effect to cause air underneath the light fixture **1** to pass through the circular channel **189**, and flow out from above the circular channel **189**; the heat generated by the light emitting plate **11** absorbed by the heat dissipating fins **188** is conducted by the heat dissipating fins **188** to the portion of the heat dissipating fins **188** located within the circular channel **189** and then carried away by the flow of air within the circular channel **189**. Because the circular channel **189** is capable of forming the chimney effect, will enhance the rate of air convection within the circular channel **189** and accelerate the rate of heat dissipation.
(60) In some embodiments, the heat sink **18** defines a through hole **187** in a center of the heat sink **18**. The lighting fixture **1** further includes a power supply **19**. The power supply **19** is mounted on a side of the mounting plate **182** back from the light emitting plate **11** and spaced from the plurality of heat dissipating fins **188**.
(61) Thereby, the flow of air from one side of the heat sink **18** to the other side of the heat sink **18** via the through hole **187** can be realized, and since the power supply **19** is mounted on the side of the mounting plate **182** back from the light emitting plate **11** and spaced apart from the plurality of heat dissipating fins **188**, the air flow space between the power supply **19** and the heat sink **18** can be increased, thereby reducing wind resistance and thus increasing the flow of air from one side of the heat sink **18** to the other side of the heat sink **18** via the through hole **187**, thereby reducing wind resistance and thereby increasing the amount of circulation flowing from one side of the heat sink **18** to the other side of the heat sink **18** via the through hole **187**.
(62) In some embodiments, the heat dissipating fins **1888** are designed with bofun type heat dissipating teeth, which can further reduce wind resistance, increase air flow, and dissipate heat better.
(63) In some embodiments, the heat dissipating fins **188** are copper structures connected to the mounting plate **182** of the heat sink **18** by soldering. It will be appreciated that in other

embodiments, the heat dissipating fins **188** are formed by aluminum extrusion in one-piece die casting together with other structural members of the heat sink **18**.
(64) In some embodiments, the heat dissipating fins **188** and the outer shell **186** are connected by means of a snap fit.
(65) In some embodiments, the light transmitting plate **13** includes a connection port **139**. The connection port **139** penetrates through the second mounting portion **132** and the gripping portion **135**. The lighting fixture **1** further includes a communication module **17**. The communication module **17** is mounted on the connection port **139**. The communication module **17** may be different types of wireless communication circuits, such as one or a combination of WIFI, infrared transceiver module, ZigBee (Purple Bee Protocol), Bluetooth. The communication module **17** may also be a microwave sensor, an infrared body detection sensor, an ambient light sensor, etc., to realize intelligent control.
(66) The above description of the technical solution of the subject matter of the present disclosure as well as the corresponding details are described above, and it can be understood that the above description is only at least one embodiment of the technical solution of the subject matter of the present disclosure, and some of the details can also be omitted in its specific implementation.
(67) In addition, in at least one embodiment of the above present disclosure, there are multiple embodiments of the combination of implementation possibilities, various combination programs are limited to space will not be listed. The person skill in the art can freely combine the implementation of the above embodiments according to the needs of the specific implementation, in order to obtain a better application experience.
(68) In summary, it can be understood that the present disclosure has the above mentioned excellent characteristics, so that it can be used to enhance the effectiveness of the previous technology has not been practical, and become a very practical value of the product.
(69) The above is only a better example of the present disclosure, and is not intended to limit the present disclosure. Any modification, equivalent substitution or improvement made within the ideas and principles of the present disclosure shall be included in the protection scope of the present disclosure.

Claims

1. A lighting fixture, wherein comprising: a lighting emitting plate; a rotating shaft being mounted on a light exiting side of the lighting emitting plate, and a light transmitting plate being located on the light exiting side of the lighting emitting plate, and being rotatably connected to the rotating shaft; light emitted from the light emitting plate coming out after passing through the light transmitting plate; wherein when the light transmitting plate is driven to make a rotating movement around a center axis of the rotating shaft, a distance between the light transmitting plate and the light emitting plate changes, and light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out changes accordingly; a circumferential wall of the rotating shaft defines a plurality of spiral grooves; the lighting fixture comprises a plurality of plug elements; the plurality of plug elements are in one-to-one correspondence with the plurality of spiral grooves; each of the plurality of plug elements comprises a first end and a second end set opposite to the first end, the first end of each plug element is connected to the light transmitting plate, the second end of each plug element is inserted into the corresponding spiral groove and is capable of sliding within the spiral groove following the rotating movement of the light transmitting plate; the spiral groove has at least two limiting points within the spiral groove; the at least two limiting points are spaced apart within the spiral groove along a spiral direction of the spiral groove; the at least two limiting points corresponds to at least two different distances between the light transmitting plate and the light emitting plate respectively, the at least two different distances between the light transmitting plate and the light emitting plate

correspond to at least two different light exiting angles respectively; and when the plug element is engaged with one of the at least two limiting points, the light transmitting plate and the light emitting plate has corresponding distance, and the light which is emitted from the light emitting plate and then passes through the light transmitting plate comes out at corresponding light exiting angle.
2. The lighting fixture according to claim 1, wherein the spiral groove comprises a spiral portion, the at least two limiting points are defined within the spiral portion.
3. The lighting fixture according to claim 2, wherein the spiral groove further comprises a transition portion and an insertion portion; the transition portion interconnects the spiral portion with the insertion portion; the rotating shaft further comprises an end face at a side away from the light emitting plate; and the insertion portion extends from the circumferential wall of the rotating shaft and penetrates through the end face of the rotating shaft.
4. The lighting fixture according to claim 3, wherein an opening of the insertion portion in the end face of the rotating shaft is larger than an outer diameter of the second end of the plug element.
5. The lighting fixture according to claim 3, wherein the spiral portion comprises a first sidewall surface close to the end face, the spiral portion comprises a stop portion formed on an end of the spiral portion close to the transition portion, the stop portion close to the opening of the insertion portion and protrudes from the first sidewall surface toward a side away from the end face.
6. The lighting fixture according to claim 3, wherein the spiral portion comprises a first sidewall surface close to the end face, the spiral portion further comprises a second sidewall surface at a side away from the end face; the second end of the plug element comprises a convex ring; and the convex ring is mated with the first sidewall surface and the second sidewall surface of the spiral portion, respectively.
7. The lighting fixture according to claim 1, wherein the light transmitting plate comprises a light transmitting portion and a mounting portion, the light transmitting portion surrounds the mounting portion, the mounting portion is recessed relative to the light transmitting portion to form a recess on a side away from the light emitting plate, an inner side wall of the recess defines a plurality of mounting holes; the first ends of the plurality of plug elements are connected within the plurality of mounting holes, respectively.
8. The lighting fixture according to claim 7, wherein the light transmitting plate comprises a light transmitting portion and a mounting portion; the light transmitting portion surrounds the mounting portion, the mounting portion is recessed relative to the light transmitting portion to form a recess on a side away from the light emitting plate; an inner side wall of the recess defines a plurality of mounting holes, the first ends of the plurality of plug elements are connected within the plurality of mounting holes, respectively.
9. The lighting fixture according to claim 1, wherein, the lighting fixture further comprises a limiting member and a heat sink, the heat sink comprises a circumferential sidewall and a mounting plate, the circumferential sidewall surrounds the mounting plate to form a mounting groove, the light emitting plate and the light transmitting plate are mounted within the mounting groove, the light transmitting plate is located on a side of the light emitting plate away from the heat sink; the limiting member is located on a side of the light transmitting plate away from the light emitting plate; one end of the limiting member is connected to the circumferential sidewall, and the other end of the limiting member is resting against a surface of the light transmitting plate away from the light emitting plate.
10. The lighting fixture according to claim 9, wherein, the light transmitting plate comprises a limiting portion at an edge of a side of the light transmitting plate away from the light emitting plate, the limiting portion cooperates with the limiting member to limit the rotating movement of the light transmitting plate relative to the light emitting plate.
11. The lighting fixture according to claim 8, wherein, the mounting portion comprises a connection port; light transmitting plate comprises a gripping portion, the connecting port

penetrates through the mounting portion and the gripping portion; the light fixture further comprises a communication module; the communication module is mounted on the connection port.

12. A lighting fixture, wherein comprising: a lighting emitting plate; a rotating shaft being mounted on a light exiting side of the lighting emitting plate, and a light transmitting plate being located on the light exiting side of the lighting emitting plate, and being rotatably connected to the rotating shaft; light emitted from the light emitting plate coming out after passing through the light transmitting plate; wherein when the light transmitting plate is driven to make a rotating movement around a center axis of the rotating shaft, a distance between the light transmitting plate and the light emitting plate changes, and light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out changes accordingly; a circumferential wall of the rotating shaft defines a plurality of spiral grooves; the lighting fixture comprises a plurality of plug elements; the plurality of plug elements are in one-to-one correspondence with the plurality of spiral grooves; each of the plurality of plug elements comprises a first end and a second end set opposite to the first end, the first end of each plug element is connected to the light transmitting plate, the second end of each plug element is inserted into the corresponding spiral groove and is capable of sliding within the spiral groove following the rotating movement of the light transmitting plate; and the light transmitting plate comprises a light transmitting portion and a mounting portion, the light transmitting portion surrounds the mounting portion, the mounting portion is recessed relative to the light transmitting portion to form a recess on a side away from the light emitting plate, an inner side wall of the recess defines a plurality of mounting holes; the first ends of the plurality of plug elements are connected within the plurality of mounting holes, respectively.

13. A lighting fixture, wherein comprising: a lighting emitting plate; a rotating shaft being mounted on a light exiting side of the lighting emitting plate, and a light transmitting plate being located on the light exiting side of the lighting emitting plate, and being rotatably connected to the rotating shaft; light emitted from the light emitting plate coming out after passing through the light transmitting plate; wherein when the light transmitting plate is driven to make a rotating movement around a center axis of the rotating shaft, a distance between the light transmitting plate and the light emitting plate changes, and light exiting angle of the light which is emitted from the light emitting plate and then passes through the light transmitting plate coming out changes accordingly; a circumferential wall of the rotating shaft defines a plurality of spiral grooves; the lighting fixture comprises a plurality of plug elements; the plurality of plug elements are in one-to-one correspondence with the plurality of spiral grooves; each of the plurality of plug elements comprises a first end and a second end set opposite to the first end, the first end of each plug element is connected to the light transmitting plate, the second end of each plug element is inserted into the corresponding spiral groove and is capable of sliding within the spiral groove following the rotating movement of the light transmitting plate; the lighting fixture further comprises a limiting member and a heat sink, the heat sink comprises a circumferential sidewall and a mounting plate, the circumferential sidewall surrounds the mounting plate to form a mounting groove, the light emitting plate and the light transmitting plate are mounted within the mounting groove, the light transmitting plate is located on a side of the light emitting plate away from the heat sink; the limiting member is located on a side of the light transmitting plate away from the light emitting plate; and one end of the limiting member is connected to the circumferential sidewall, and the other end of the limiting member is resting against a surface of the light transmitting plate away from the light emitting plate.